0.1 Introduction

I have always been a little fascinated by flight crews: the idea of being paid to fly somewhere and then getting to travel once you land (at least that is what I imagined the job to be). That curiosity is what drew me to this report, which aims to reproduce and potentially improve upon the results of the paper "Individual scheduling approach for multi-class airline cabin crew with manpower requirement heterogeneity"[1] [1]. This report focuses on the integer programming aspects of the paper.

0.1.1 Background Problem

The paper tackles the cabin crew pairing problem (CPP): building legal sequences of flights (pairings) that start and end at a home base and cover every flight, while minimising cost. Crew cost is an airline's second-largest expense after fuel, so small scheduling gains matter. Cockpit crews are qualified for one aircraft type with fixed requirements and are scheduled as teams. Cabin crews are different: they are split into multiple classes, are cross-qualified across aircraft types, and the number of each class required varies by aircraft, cabin layout, and flight. This heterogeneity is the core difficulty.

Modelling cabin crews as teams is wasteful here. Since a team flies together, its size must meet the maximum requirement across all flights on the pairing, leaving surplus crew idle on lighter flights (manpower waste). Scheduling crew individually avoids this and raises utilisation. The paper also formalises Controlled Crew Substitution (CCS): when flight fluctuation leaves a class short, a crew member of another class temporarily covers the duty, so a flight is fine as long as total crew meets total demand. Combining these ideas gives the proposed MICCPP-ACCS model, which schedules each crew individually, adds per-class availability limits, and embeds CCS.

0.1.2 Original Paper's methodology

The paper builds three integer programs. TCCPP is the traditional team-based set-covering model minimising time-away-from-base (TAFB) with no availability limit. MICCPP-ACCS is the proposed individual model whose objective orders pairing cost, a substitution penalty, and a heavily penalised extra-crew cost so that available crew are used first, CCS only on real shortage, and extra crew last; its constraints handle total satisfaction, per-class minimum satisfaction, substitution recording, and availability. MICCPP-A removes CCS and is used to derive three benchmarks (MS, MC_r, MM_r) that feed an eight-scenario case analysis of when CCS or extra crew are needed. Solutions come from a column generation plus MIP heuristic (with a DPIA initialiser and a labelling algorithm for the pricing problem), and from a Genetic Algorithm on large instances. Experiments on a Hong Kong–Singapore route and on large hypothetical instances show MICCPP-ACCS eliminates idle crews, lowers cost against both traditional variants, and uses CCS to cut extra-manpower demand.

0.1.3 Limitations of the original models

Two modelling choices stand out as limitations. First, the formulation plans around a single home base (Hong Kong): availability is just a cap on pairings per class, and any shortage is filled by "extra" crew that the model can effectively import from anywhere at a fixed penalty. This sidesteps the real geography of crew bases, positioning, and where spare crew actually are, so the availability constraint is a rough proxy rather than a true resourcing model. Second, the model lets any class substitute any other class. If substitution is unrestricted in both directions, it is unclear what the class distinction is buying. In practice higher classes substitute lower ones, not the reverse; allowing free two-way substitution risks overstating the flexibility (and the savings) that CCS delivers. Third, cost is approximated purely by time-away-from-base, treating every minute of a pairing as equal. In reality the segments of a duty cost differently: active flight time, ground waiting (sit) time, and deadhead repositioning are not interchangeable, so a single TAFB figure can misrank pairings that an airline would price very differently.

[1] X. Wen, S.-H. Chung, P. Ji, and J.-B. Sheu, *Transportation Research Part E*, vol. 163, 102763, 2022.